Lesson 12: Inequalities and Limit Theorems

AI/ML Mathematics

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1 Motivation and Overview

Suppose a reinforcement learning agent uses Monte Carlo rollouts to estimate the value of a policy: it runs n independent episodes starting from a fixed state s, collects the discounted returns $G^{(1)}, \ldots, G^{(n)}$, and forms the sample mean $\bar{G}_n = \frac{1}{n}\sum_{i=1}^{n} G^{(i)}$. The true value $V^\pi(s) = \mathbb{E}[G^{(i)}]$ is unknown. Three fundamental questions arise:

1. Does $\bar{G}_n$ converge to $V^\pi(s)$ as $n \to \infty$, and in what sense?
2. How quickly does $\bar{G}_n$ concentrate around $V^\pi(s)$ – that is, how large must n be so that $|\bar{G}_n - V^\pi(s)| < \varepsilon$ with probability at least $1 - \delta$?
3. What is the approximate distribution of the estimation error $\bar{G}_n - V^\pi(s)$ for large n?

The first question is answered by the law of large numbers, the second by concentration inequalities (Chebyshev and Hoeffding), and the third by the central limit theorem. The tools developed in this lesson will let us prove these results, derive explicit sample complexity bounds for Monte Carlo estimation, and understand why the normal distribution appears as a universal limit.

Why Inequalities and Limit Theorems Matter for RL/ML

- **Monte Carlo policy evaluation**: The law of large numbers guarantees that averaging returns converges to the true value function; concentration inequalities quantify how many episodes are needed for a given accuracy.
- **TD-learning convergence rates**: Stochastic approximation convergence proofs rely on Markov's and Chebyshev's inequalities to control the error at each iteration (Phase 01, Lesson on Stochastic Approximation).
- **PAC-MDP bounds**: Sample complexity results in PAC-MDP theory use Hoeffding's inequality to bound the probability of inaccurate value estimates.
- **Bandit confidence intervals**: UCB algorithms construct confidence bounds for arm means using Hoeffding's inequality; the central limit theorem justifies Gaussian approximations for large sample sizes.

Prerequisites. Probability spaces and the Kolmogorov axioms (Lesson 9). Random variables, PMFs, PDFs, CDFs, and standard distributions – in particular the Bernoulli, Poisson, and Normal distributions (Lesson 10). Expectation, variance, covariance, independence of random variables, the law of the unconscious statistician (LOTUS), linearity of expectation, and the tower property (Lesson 11).

Learning objectives. After completing this lesson, the student will be able to:

- *Define* the moment generating function and *derive* moments from it by differentiation.
- *Prove* Markov's, Chebyshev's, Cauchy-Schwarz, and Jensen's inequalities.
- *State* the definitions of convergence in probability and convergence in distribution, and *prove* the weak law of large numbers via Chebyshev's inequality.
- *State* the strong law of large numbers and the central limit theorem, and *apply* them in worked examples.
- *Prove* Hoeffding's lemma and Hoeffding's inequality, and *derive* sample complexity bounds for Monte Carlo estimation.
- *Compare* the sample complexity guarantees provided by Chebyshev's inequality versus Hoeffding's inequality.

2 References

- Ross [2019, Chapters 7–8] – Moment generating functions, limit theorems, and concentration inequalities.
- Blitzstein and Hwang [2019, Chapter 10] – Inequalities and limit theorems with extensive worked examples and RL-relevant applications.
- Feller [1968, Chapters 8–10] – Classical treatment of the law of large numbers and the central limit theorem.
- Durrett [2019, Sections 2.2–2.3] – Rigorous treatment of the weak and strong laws of large numbers and the central limit theorem via characteristic functions.
- Grimmett and Stirzaker [2020, Sections 5.1–5.4] – Generating functions, moment generating functions, and convergence of random variables.

3 Moment Generating Functions

Moment generating functions (MGFs) provide a powerful technique for computing moments, identifying distributions, and establishing the distribution of sums of independent random variables.

Definition 3.1 (Moment Generating Function)**.** Let X be a random variable. The *moment generating function* (MGF) of X is the function $M_X \colon \mathbb{R} \to [0, \infty]$ defined by

$$M_X(t) = \mathbb{E}[e^{tX}],$$

provided the expectation exists. We say the MGF exists in a neighborhood of the origin if there exists $h > 0$ such that $M_X(t) < \infty$ for all $|t| < h$.

Remark 3.2. Note that $M_X(0) = \mathbb{E}[e^0] = 1$ always holds. The MGF may fail to exist for $t \neq 0$; for instance, the Cauchy distribution has no finite moments beyond order zero, so its MGF is infinite for all $t \neq 0$. When we say "the MGF exists," we mean it is finite in some open interval containing the origin.

Theorem 3.3 (Moments from the MGF)**.** *If the MGF $M_X(t)$ exists in a neighborhood of the origin, then $\mathbb{E}[|X|^n] < \infty$ for all $n \geq 1$ and*

$$\mathbb{E}[X^n] = M_X^{(n)}(0),$$

where $M_X^{(n)}$ denotes the n-th derivative of M_X.

Proof. Since $M_X(t) = \mathbb{E}[e^{tX}]$ is finite for $|t| < h$, we can differentiate under the expectation sign (justified by the dominated convergence theorem, since $|X^n e^{tX}|$ is dominated by a sum of exponentials that are integrable for $|t|$ sufficiently small). We have

$$M_X'(t) = \frac{d}{dt}\mathbb{E}[e^{tX}] = \mathbb{E}\left[\frac{d}{dt}e^{tX}\right] = \mathbb{E}[Xe^{tX}].$$

Evaluating at $t = 0$: $M_X'(0) = \mathbb{E}[X \cdot 1] = \mathbb{E}[X]$.

More generally, by induction, the n-th derivative is

$$M_X^{(n)}(t) = \mathbb{E}[X^n e^{tX}],$$

and evaluating at $t = 0$ gives $M_X^{(n)}(0) = \mathbb{E}[X^n]$.

For the base case $n = 1$, we have shown $M'_X(t) = \mathbb{E}[Xe^{tX}]$. Assume $M_X^{(k)}(t) = \mathbb{E}[X^k e^{tX}]$ for some $k \geq 1$. Then

$$M_X^{(k+1)}(t) = \frac{d}{dt}\mathbb{E}[X^k e^{tX}] = \mathbb{E}\left[X^k \frac{d}{dt} e^{tX}\right] = \mathbb{E}[X^{k+1} e^{tX}],$$

completing the induction. □

Example 3.4 (First Two Moments via the MGF)**.** The first two moments are $\mathbb{E}[X] = M'_X(0)$ and $\mathbb{E}[X^2] = M''_X(0)$. The variance can then be computed as

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = M''_X(0) - (M'_X(0))^2.$$

3.1 MGF of a Sum of Independent Random Variables

Theorem 3.5 (MGF of a Sum of Independents)**.** *If X and Y are independent random variables whose MGFs exist, then*

$$M_{X+Y}(t) = M_X(t) \cdot M_Y(t).$$

More generally, if $X_1, \ldots, X_n$ are independent, then $M_{X_1+\cdots+X_n}(t) = \prod_{i=1}^n M_{X_i}(t)$.

Proof. By definition,

$$M_{X+Y}(t) = \mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX} e^{tY}].$$

Since X and Y are independent, e^{tX} and e^{tY} are also independent (as they are functions of independent random variables; see Lesson 11, Proposition **??**). Therefore

$$\mathbb{E}[e^{tX} e^{tY}] = \mathbb{E}[e^{tX}] \cdot \mathbb{E}[e^{tY}] = M_X(t) \cdot M_Y(t).$$

The general case follows by induction on n. □

Remark 3.6. A fundamental result from the theory of Laplace transforms (see, e.g., Grimmett and Stirzaker [2020, Theorem 2.3.6]) states that if the MGF of X exists in a neighborhood of the origin, then it uniquely determines the distribution of X. That is, if $M_X(t) = M_Y(t)$ for all t in some interval $(-h, h)$ with $h > 0$, then X and Y have the same distribution. We state this without proof, as it requires complex analysis (the inversion formula for Laplace transforms).

3.2 MGFs of Standard Distributions

Proposition 3.7 (MGF of the Bernoulli Distribution)**.** *If $X \sim \mathrm{Bernoulli}(p)$, then*

$$M_X(t) = 1 - p + pe^t \quad \textit{for all } t \in \mathbb{R}.$$

Proof. Since X takes value 1 with probability p and 0 with probability $1 - p$,

$$M_X(t) = \mathbb{E}[e^{tX}] = e^{t \cdot 0}(1 - p) + e^{t \cdot 1} p = (1 - p) + pe^t.$$ □

Proposition 3.8 (MGF of the Poisson Distribution)**.** *If $X \sim \mathrm{Poisson}(\lambda)$, then*

$$M_X(t) = e^{\lambda(e^t - 1)} \quad \textit{for all } t \in \mathbb{R}.$$

Proof. Using the PMF $\mathbb{P}(X = k) = e^{-\lambda}\lambda^k/k!$,

$$M_X(t) = \mathbb{E}[e^{tX}] = \sum_{k=0}^{\infty} e^{tk} \frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} \cdot e^{\lambda e^t} = e^{\lambda(e^t - 1)},$$

where we recognized the Taylor series $\sum_{k=0}^{\infty} z^k/k! = e^z$ with $z = \lambda e^t$. □

Proposition 3.9 (MGF of the Normal Distribution)**.** *If $X \sim \mathcal{N}(\mu, \sigma^2)$, then*

$$M_X(t) = e^{\mu t + \sigma^2 t^2/2} \quad \textit{for all } t \in \mathbb{R}.$$

Proof. First consider $Z \sim \mathcal{N}(0,1)$. Then

$$M_Z(t) = \mathbb{E}[e^{tZ}] = \int_{-\infty}^{\infty} e^{tz} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz - z^2/2}\, dz.$$

Complete the square in the exponent: $tz - z^2/2 = -(z^2 - 2tz)/2 = -((z-t)^2 - t^2)/2 = -(z-t)^2/2 + t^2/2$. Therefore

$$M_Z(t) = \frac{e^{t^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(z-t)^2/2}\, dz = e^{t^2/2},$$

since $\int_{-\infty}^{\infty} e^{-(z-t)^2/2}\, dz = \sqrt{2\pi}$ by the Gaussian integral (Lesson 10) with a shift of variable $u = z - t$.

For general $X = \mu + \sigma Z$ where $Z \sim \mathcal{N}(0,1)$,

$$M_X(t) = \mathbb{E}[e^{t(\mu + \sigma Z)}] = e^{\mu t}\mathbb{E}[e^{(t\sigma)Z}] = e^{\mu t} M_Z(t\sigma) = e^{\mu t} e^{\sigma^2 t^2/2} = e^{\mu t + \sigma^2 t^2/2}.$$ □

3.3 Applications of the MGF

Example 3.10 (Sum of Independent Poissons)**.** Let $X_1 \sim \text{Poisson}(\lambda_1)$ and $X_2 \sim \text{Poisson}(\lambda_2)$ be independent. By Theorem 3.5,

$$M_{X_1+X_2}(t) = M_{X_1}(t) M_{X_2}(t) = e^{\lambda_1(e^t - 1)} e^{\lambda_2(e^t-1)} = e^{(\lambda_1+\lambda_2)(e^t-1)}.$$

This is the MGF of $\text{Poisson}(\lambda_1 + \lambda_2)$. By the uniqueness of the MGF (Remark 3.6), $X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

Example 3.11 (Sum of Independent Normals)**.** Let $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ be independent. Then

$$M_{X_1+X_2}(t) = e^{\mu_1 t + \sigma_1^2 t^2/2} \cdot e^{\mu_2 t + \sigma_2^2 t^2/2} = e^{(\mu_1+\mu_2)t + (\sigma_1^2+\sigma_2^2)t^2/2}.$$

This is the MGF of $\mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. By uniqueness, $X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

MGFs and Reward Distributions

In RL, rewards are often modeled as sums of independent random variables. The MGF factorization theorem (Theorem 3.5) immediately identifies the distribution of cumulative rewards when individual rewards are independent. For instance, if each of n independent episodes yields a Poisson-distributed count of some event with parameter λ, the total count across all episodes is $\text{Poisson}(n\lambda)$. Similarly, if reward noise is Gaussian, the average reward over n episodes remains Gaussian with variance shrinking as $1/n$.

We now turn to probability inequalities, which provide deterministic upper bounds on tail probabilities without requiring knowledge of the full distribution.

4 Probability Inequalities

Probability inequalities bound the probability that a random variable deviates from its mean. These bounds are essential for establishing convergence rates and deriving sample complexity results.

4.1 Markov's Inequality

Theorem 4.1 (Markov's Inequality)**.** *Let X be a non-negative random variable with finite expectation. Then for any $a > 0$,*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof. Define the indicator $\mathbf{1}_{\{X \geq a\}}$. Since $X \geq 0$, we have the pointwise inequality

$$X \geq a \cdot \mathbf{1}_{\{X \geq a\}},$$

because when $X \geq a$ the right side equals $a \leq X$, and when $X < a$ the right side equals $0 \leq X$. Taking expectations and using linearity:

$$\mathbb{E}[X] \geq a \cdot \mathbb{E}[\mathbf{1}_{\{X \geq a\}}] = a \cdot \mathbb{P}(X \geq a).$$

Dividing both sides by $a > 0$ gives $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$. □

Example 4.2 (Markov Applied to a Uniform Variable)**.** Let $X \sim \text{Uniform}(0, 1)$, so $\mathbb{E}[X] = 1/2$. For $a = 3/4$, Markov gives $\mathbb{P}(X \geq 3/4) \leq (1/2)/(3/4) = 2/3$. The exact probability is $\mathbb{P}(X \geq 3/4) = 1/4$, so the bound is loose but valid. Markov's inequality uses only the mean, so its generality comes at the cost of tightness.

4.2 Chebyshev's Inequality

Theorem 4.3 (Chebyshev's Inequality)**.** *Let X be a random variable with finite mean $\mu = \mathbb{E}[X]$ and finite variance $\sigma^2 = \text{Var}(X)$. Then for any $k > 0$,*

$$\mathbb{P}(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

Proof. Apply Markov's inequality (Theorem 4.1) to the non-negative random variable $Y = (X - \mu)^2$ with threshold $a = k^2$:

$$\mathbb{P}(|X - \mu| \geq k) = \mathbb{P}((X - \mu)^2 \geq k^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{k^2} = \frac{\text{Var}(X)}{k^2} = \frac{\sigma^2}{k^2}.$$ □

Example 4.4 (Chebyshev for a Poisson Variable)**.** Let $X \sim \text{Poisson}(\lambda)$ with $\mathbb{E}[X] = \lambda$ and $\text{Var}(X) = \lambda$ (Lesson 10). Chebyshev gives

$$\mathbb{P}(|X - \lambda| \geq k) \leq \frac{\lambda}{k^2}.$$

For $\lambda = 100$ and $k = 20$: $\mathbb{P}(|X - 100| \geq 20) \leq 100/400 = 1/4$.

Example 4.5 (Chebyshev for a Sample Mean)**.** Let $X_1, \ldots, X_n$ be i.i.d. with mean μ and variance σ^2. The sample mean $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ has $\mathbb{E}[\bar{X}_n] = \mu$ and $\text{Var}(\bar{X}_n) = \sigma^2/n$ (by independence and the variance of a sum; see Lesson 11, Corollary 4.9). Chebyshev gives

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

This bound decreases as $O(1/n)$, confirming that sample means concentrate around the population mean.

4.3 Cauchy-Schwarz Inequality for Expectations

Theorem 4.6 (Cauchy-Schwarz Inequality). *For any random variables X and Y with finite second moments,*

$$|\mathbb{E}[XY]|^2 \leq \mathbb{E}[X^2]\,\mathbb{E}[Y^2].$$

Equality holds if and only if $Y = cX$ almost surely for some constant $c \in \mathbb{R}$ (or $X = 0$ a.s.).

Proof. If $\mathbb{E}[X^2] = 0$, then $X = 0$ a.s., so $\mathbb{E}[XY] = 0$ and the inequality holds trivially. Assume $\mathbb{E}[X^2] > 0$.

For any $t \in \mathbb{R}$, the random variable $(tX + Y)^2$ is non-negative, so

$$0 \leq \mathbb{E}[(tX + Y)^2] = t^2\mathbb{E}[X^2] + 2t\mathbb{E}[XY] + \mathbb{E}[Y^2].$$

This is a quadratic in t that is non-negative for all t. A real quadratic $at^2 + bt + c \geq 0$ for all t if and only if its discriminant satisfies $b^2 - 4ac \leq 0$. Here $a = \mathbb{E}[X^2]$, $b = 2\mathbb{E}[XY]$, $c = \mathbb{E}[Y^2]$, so

$$4(\mathbb{E}[XY])^2 - 4\mathbb{E}[X^2]\mathbb{E}[Y^2] \leq 0,$$

which gives $(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]$.

Equality holds if and only if the discriminant is zero, meaning the quadratic has a real root t_0 with $\mathbb{E}[(t_0X + Y)^2] = 0$, so $Y = -t_0X$ almost surely. □

Remark 4.7. The Cauchy-Schwarz inequality immediately implies the correlation bound $|\rho(X, Y)| \leq 1$ (Lesson 11, Proposition 4.14). It also appears in Phase 01 (Lesson 7) as a fundamental property of inner product spaces, where $\mathbb{E}[XY]$ plays the role of the inner product on L^2.

4.4 Jensen's Inequality

Definition 4.8 (Convex Function). A function $g\colon \mathbb{R} \to \mathbb{R}$ is *convex* if for all $x, y \in \mathbb{R}$ and $\lambda \in [0, 1]$,

$$g(\lambda x + (1 - \lambda)y) \leq \lambda g(x) + (1 - \lambda)g(y).$$

Theorem 4.9 (Jensen's Inequality). *Let X be a random variable with finite expectation and $g\colon \mathbb{R} \to \mathbb{R}$ a convex function. Then*

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)],$$

provided $\mathbb{E}[|g(X)|] < \infty$.

Proof. We prove the result for a discrete random variable X taking finitely many values, using induction on the number of values.

Base case ($n = 1$): If $X = x_1$ with probability 1, then $g(\mathbb{E}[X]) = g(x_1) = \mathbb{E}[g(X)]$, so equality holds.

Base case ($n = 2$): If X takes values x_1, x_2 with probabilities p_1, p_2 where $p_1 + p_2 = 1$, then $\mathbb{E}[X] = p_1x_1 + p_2x_2$ and

$$g(\mathbb{E}[X]) = g(p_1x_1 + p_2x_2) \leq p_1g(x_1) + p_2g(x_2) = \mathbb{E}[g(X)],$$

by the definition of convexity with $\lambda = p_1$.

Inductive step: Assume the result holds for random variables taking at most $n - 1$ values. Suppose X takes values $x_1, \ldots, x_n$ with probabilities $p_1, \ldots, p_n$, where $p_n < 1$ (otherwise X is constant and the result is trivial). Let $q = 1 - p_n > 0$. Then

$$\mathbb{E}[X] = \sum_{i=1}^{n} p_ix_i = q\underbrace{\left(\sum_{i=1}^{n-1} \frac{p_i}{q}x_i\right)}_{=:\mu} + p_nx_n.$$

Since $\sum_{i=1}^{n-1} p_i/q = 1$, the quantity μ is the expectation of a random variable Y taking values $x_1, \ldots, x_{n-1}$ with probabilities $p_1/q, \ldots, p_{n-1}/q$. By convexity of g (with $\lambda = q$):

$$g(\mathbb{E}[X]) = g(q\mu + p_n x_n) \leq q\, g(\mu) + p_n\, g(x_n).$$

By the inductive hypothesis applied to Y: $g(\mu) = g(\mathbb{E}[Y]) \leq \mathbb{E}[g(Y)] = \sum_{i=1}^{n-1}(p_i/q)g(x_i)$. Therefore

$$g(\mathbb{E}[X]) \leq q \sum_{i=1}^{n-1} \frac{p_i}{q} g(x_i) + p_n g(x_n) = \sum_{i=1}^{n} p_i g(x_i) = \mathbb{E}[g(X)]. \qquad \square$$

Example 4.10 (Jensen Applied to Common Functions). (a) Since $g(x) = x^2$ is convex, Jensen gives $(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2]$, which is equivalent to $\mathrm{Var}(X) \geq 0$.

(b) Since $g(x) = e^x$ is convex, Jensen gives $e^{\mathbb{E}[X]} \leq \mathbb{E}[e^X] = M_X(1)$.

(c) Since $g(x) = -\ln x$ is convex on $(0, \infty)$, Jensen gives $-\ln(\mathbb{E}[X]) \leq \mathbb{E}[-\ln X]$, i.e., $\ln(\mathbb{E}[X]) \geq \mathbb{E}[\ln X]$. This is the AM-GM inequality in its probabilistic form.

Probability Inequalities Summary

- **Markov**: $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$ for $X \geq 0$, $a > 0$. Uses only the first moment.
- **Chebyshev**: $\mathbb{P}(|X - \mu| \geq k) \leq \sigma^2/k^2$. Uses first two moments.
- **Cauchy-Schwarz**: $|\mathbb{E}[XY]|^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]$.
- **Jensen**: $g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$ for convex g.

These inequalities form a hierarchy: Chebyshev follows from Markov, and each successive result incorporates more information about the distribution, yielding tighter bounds.

Concentration Inequalities and Convergence Rates in RL

Concentration inequalities translate directly into convergence rate guarantees for RL algorithms. Chebyshev's inequality applied to the sample mean gives an $O(1/n)$ bound on the probability of ε-deviation, leading to $O(1/(\varepsilon^2\delta))$ sample complexity (Section 7). Exponential inequalities such as Hoeffding's (Section 7) improve this to $O(\ln(1/\delta)/\varepsilon^2)$, which is the standard dependence in PAC-MDP theory and bandit algorithms.

With the fundamental inequalities in hand, we now study the asymptotic behavior of sample means through the law of large numbers and the central limit theorem.

5 Convergence Concepts and the Law of Large Numbers

5.1 Modes of Convergence

Definition 5.1 (Convergence in Probability). A sequence of random variables $(X_n)_{n\geq 1}$ *converges in probability* to a random variable X, written $X_n \xrightarrow{\mathbb{P}} X$, if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|X_n - X| \geq \varepsilon) = 0.$$

Definition 5.2 (Almost Sure Convergence). A sequence $(X_n)_{n\geq 1}$ *converges almost surely* to X, written $X_n \xrightarrow{\text{a.s.}} X$, if

$$\mathbb{P}\Big(\lim_{n\to\infty} X_n = X\Big) = 1.$$

Definition 5.3 (Convergence in Distribution). A sequence $(X_n)_{n\geq 1}$ *converges in distribution* to X, written $X_n \xrightarrow{d} X$, if

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$$

at every point x where F_X is continuous. Here F_{X_n} and F_X are the CDFs of X_n and X, respectively.

Remark 5.4. Almost sure convergence implies convergence in probability, which in turn implies convergence in distribution:

$$X_n \xrightarrow{\text{a.s.}} X \implies X_n \xrightarrow{\mathbb{P}} X \implies X_n \xrightarrow{d} X.$$

The converses are generally false. We will not prove these implications here; they are established rigorously in Phase 01 (measure-theoretic probability).

5.2 Weak Law of Large Numbers

Theorem 5.5 (Weak Law of Large Numbers (WLLN)). *Let $X_1, X_2, \ldots$ be i.i.d. random variables with finite mean $\mu = \mathbb{E}[X_1]$ and finite variance $\sigma^2 = \mathrm{Var}(X_1)$. Define $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$. Then*

$$\bar{X}_n \xrightarrow{\mathbb{P}} \mu,$$

i.e., for every $\varepsilon > 0$, $\lim_{n\to\infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) = 0$.

Proof. By Example 4.5 (Chebyshev's inequality applied to $\bar{X}_n$),

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}.$$

For fixed $\varepsilon > 0$, the right side tends to 0 as $n \to \infty$. Therefore $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$. □

The Law of Large Numbers

The weak law of large numbers guarantees that sample averages converge in probability to the population mean. This is the theoretical foundation for Monte Carlo estimation: averaging i.i.d. samples from any distribution with finite variance yields a consistent estimator of the mean. In RL, this justifies computing $V^\pi(s) \approx \frac{1}{n}\sum_{i=1}^n G^{(i)}$ where the $G^{(i)}$ are i.i.d. returns from state s.

Example 5.6 (WLLN for a Biased Coin). Toss a coin with $\mathbb{P}(\text{heads}) = p$ independently n times. Let $X_i = \mathbf{1}_{\{\text{heads on toss } i\}}$, so each $X_i \sim \text{Bernoulli}(p)$ with $\mu = p$ and $\sigma^2 = p(1-p) \leq 1/4$. The sample mean $\bar{X}_n = (\text{number of heads})/n$ is the empirical frequency of heads. By the WLLN, $\bar{X}_n \xrightarrow{\mathbb{P}} p$: the observed frequency converges (in probability) to the true probability. Chebyshev gives the quantitative bound $\mathbb{P}(|\bar{X}_n - p| \geq \varepsilon) \leq 1/(4n\varepsilon^2)$.

5.3 Strong Law of Large Numbers

Theorem 5.7 (Strong Law of Large Numbers (SLLN)). *Let $X_1, X_2, \ldots$ be i.i.d. random variables with finite mean $\mu = \mathbb{E}[X_1]$. Then*

$$\bar{X}_n \xrightarrow{a.s.} \mu.$$

The SLLN strengthens the WLLN in two ways: it requires only finite mean (not finite variance), and it gives almost sure convergence rather than convergence in probability. The proof requires measure-theoretic tools – specifically the Borel-Cantelli lemma and truncation arguments – and is deferred to Phase 01 (Lesson 5, probability foundations).

WLLN vs. SLLN: Finite Variance is Not Always Needed

The WLLN proof above requires finite variance. This is a limitation of the Chebyshev-based approach, not of the WLLN itself. A more refined proof using truncation (see Durrett [2019, Section 2.2]) establishes the WLLN under the weaker condition $\mathbb{E}[|X_1|] < \infty$ alone. The SLLN also requires only $\mathbb{E}[|X_1|] < \infty$. For the purpose of this course, we state the SLLN without proof and note that in RL applications, bounded rewards automatically guarantee finite moments of all orders.

6 The Central Limit Theorem

The law of large numbers tells us that $\bar{X}_n \to \mu$; the central limit theorem describes the *rate* and *shape* of the fluctuations around μ.

Theorem 6.1 (Central Limit Theorem (CLT))**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables with finite mean $\mu = \mathbb{E}[X_1]$ and finite variance $\sigma^2 = \mathrm{Var}(X_1) > 0$. Then*

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1),$$

i.e., $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ converges in distribution to a standard normal random variable.

The proof uses characteristic functions (or equivalently, MGFs when they exist) and is deferred to Phase 01. We focus here on understanding and applying the CLT.

Remark 6.2. The CLT says that for large n, the sample mean is approximately normal:

$$\bar{X}_n \dot{\sim} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

This approximation improves as n increases and holds regardless of the distribution of the X_i (provided σ^2 is finite).

Example 6.3 (CLT for Polling)**.** A pollster surveys $n = 1000$ voters, each independently supporting candidate A with probability $p = 0.52$. Let $\hat{p} = \bar{X}_n$ be the sample proportion. By the CLT,

$$\hat{p} \dot{\sim} \mathcal{N}\left(0.52, \frac{0.52 \times 0.48}{1000}\right) = \mathcal{N}(0.52, 0.0002496).$$

The standard deviation of $\hat{p}$ is $\sigma/\sqrt{n} = \sqrt{0.0002496} \approx 0.0158$. The probability that the poll shows candidate A trailing ($\hat{p} < 0.5$) is approximately

$$\mathbb{P}(\hat{p} < 0.5) \approx \mathbb{P}\left(Z < \frac{0.5 - 0.52}{0.0158}\right) = \mathbb{P}(Z < -1.27) \approx 0.102,$$

where $Z \sim \mathcal{N}(0,1)$.

Example 6.4 (CLT for Dice Sums)**.** Roll a fair six-sided die $n = 100$ times and let $S_{100} = \sum_{i=1}^{100} X_i$ be the total. Each X_i has $\mu = 7/2$ and $\sigma^2 = 35/12$ (Lesson 10). By the CLT,

$$S_{100} \dot{\sim} \mathcal{N}(350, 100 \cdot 35/12) = \mathcal{N}(350, 291.67).$$

The probability that S_{100} exceeds 375 is approximately

$$\mathbb{P}(S_{100} > 375) \approx \mathbb{P}\left(Z > \frac{375 - 350}{\sqrt{291.67}}\right) = \mathbb{P}(Z > 1.464) \approx 0.072.$$

Despite the discrete, non-Gaussian distribution of each X_i, the sum is well-approximated by a normal distribution.

CLT and Confidence Intervals for Policy Evaluation

In Monte Carlo policy evaluation, the CLT provides confidence intervals for the estimated value function. If $G^{(1)}, \ldots, G^{(n)}$ are i.i.d. returns with mean $V^\pi(s)$ and variance σ^2, then by the CLT an approximate 95% confidence interval for $V^\pi(s)$ is

$$\bar{G}_n \pm 1.96 \cdot \frac{\hat{\sigma}}{\sqrt{n}},$$

where $\hat{\sigma}$ is the sample standard deviation. This is how practitioners assess the precision of value estimates and decide when to stop collecting episodes.

7 Tail Bounds and Sample Complexity

The Chebyshev bound provides a polynomial $(1/n)$ rate of concentration. For bounded random variables, we can obtain much tighter exponential bounds via the Chernoff method and Hoeffding's inequality.

7.1 Chebyshev Confidence Interval

Before developing Hoeffding's inequality, let us formalize the sample complexity implied by Chebyshev's inequality.

Proposition 7.1 (Chebyshev Sample Complexity)**.** *Let $X_1, \ldots, X_n$ be i.i.d. with mean μ and variance σ^2. To ensure $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \delta$, it suffices to take*

$$n \geq \frac{\sigma^2}{\varepsilon^2 \delta}.$$

Proof. From Chebyshev's inequality (Example 4.5), $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2)$. Setting $\sigma^2/(n\varepsilon^2) \leq \delta$ and solving for n gives $n \geq \sigma^2/(\varepsilon^2\delta)$. □

Example 7.2 (Chebyshev Sample Size)**.** For i.i.d. Bernoulli rewards with $\sigma^2 \leq 1/4$, to estimate the mean within $\varepsilon = 0.05$ with probability at least $1 - \delta = 0.95$ ($\delta = 0.05$), Chebyshev requires

$$n \geq \frac{1/4}{0.05^2 \cdot 0.05} = \frac{0.25}{0.000125} = 2000.$$

7.2 The Chernoff Method

The key idea for obtaining exponential tail bounds is the *Chernoff method*: apply Markov's inequality to e^{tX} for an optimally chosen $t > 0$.

Proposition 7.3 (Chernoff Bound – Upper Tail)**.** *For any random variable X and $a \in \mathbb{R}$,*

$$\mathbb{P}(X \geq a) \leq \inf_{t>0} e^{-ta} M_X(t),$$

provided the MGF $M_X(t)$ exists.

Proof. For any $t > 0$, the event $\{X \geq a\}$ is equivalent to $\{e^{tX} \geq e^{ta}\}$. Since $e^{tX} \geq 0$, Markov's inequality gives

$$\mathbb{P}(X \geq a) = \mathbb{P}(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}} = e^{-ta} M_X(t).$$

This holds for every $t > 0$, so we can take the infimum over $t > 0$. □

7.3 Hoeffding's Lemma

Lemma 7.4 (Hoeffding's Lemma)**.** *Let X be a random variable with $\mathbb{E}[X] = 0$ and $a \le X \le b$ almost surely. Then for all $t \in \mathbb{R}$,*

$$\mathbb{E}[e^{tX}] \le \exp\left(\frac{t^2(b-a)^2}{8}\right).$$

Proof. Since $a \le X \le b$, the random variable X can be written as a convex combination of the endpoints. For any $x \in [a, b]$,

$$x = \frac{b-x}{b-a} \cdot a + \frac{x-a}{b-a} \cdot b.$$

Since e^{tx} is convex in x (it is the exponential function composed with a linear function), we have

$$e^{tx} \le \frac{b-x}{b-a} e^{ta} + \frac{x-a}{b-a} e^{tb}.$$

Taking expectations (and using $\mathbb{E}[X] = 0$):

$$\begin{aligned} \mathbb{E}[e^{tX}] &\le \frac{b-\mathbb{E}[X]}{b-a} e^{ta} + \frac{\mathbb{E}[X]-a}{b-a} e^{tb} \\ &= \frac{b}{b-a} e^{ta} + \frac{-a}{b-a} e^{tb}. \end{aligned} \tag{1}$$

Define $p = -a/(b-a)$ (note $p \in [0, 1]$ since $a \le 0 \le b$, which follows from $\mathbb{E}[X] = 0$ and $a \le X \le b$) and $u = t(b-a)$. Then (1) becomes

$$\mathbb{E}[e^{tX}] \le (1-p)e^{-pu} + p\, e^{(1-p)u} = e^{\varphi(u)},$$

where $\varphi(u) = -pu + \ln(1 - p + pe^u)$.

We claim that $\varphi(u) \le u^2/8$ for all $u \in \mathbb{R}$. To verify this, note $\varphi(0) = 0$ and $\varphi'(0) = 0$:

$$\varphi'(u) = -p + \frac{pe^u}{1-p+pe^u}, \qquad \varphi'(0) = -p + \frac{p}{1} = 0.$$

The second derivative is

$$\varphi''(u) = \frac{pe^u(1-p+pe^u) - (pe^u)^2}{(1-p+pe^u)^2} = \frac{pe^u(1-p)}{(1-p+pe^u)^2}.$$

Let $q = pe^u/(1-p+pe^u) \in (0, 1)$. Then $\varphi''(u) = q(1-q) \le 1/4$ (since $q(1-q)$ achieves its maximum of $1/4$ at $q = 1/2$).

By Taylor's theorem with Lagrange remainder, there exists ξ between 0 and u such that

$$\varphi(u) = \varphi(0) + \varphi'(0)u + \frac{1}{2}\varphi''(\xi)u^2 = \frac{1}{2}\varphi''(\xi)u^2 \le \frac{u^2}{8}.$$

Since $u = t(b-a)$, we conclude $\mathbb{E}[e^{tX}] \le e^{t^2(b-a)^2/8}$. □

7.4 Hoeffding's Inequality

Theorem 7.5 (Hoeffding's Inequality)**.** *Let $X_1, \ldots, X_n$ be independent random variables with $a_i \le X_i \le b_i$ almost surely for each i. Let $S_n = \sum_{i=1}^n X_i$ and $\mu = \mathbb{E}[S_n]$. Then for any $t > 0$,*

$$\mathbb{P}(S_n - \mu \ge t) \le \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

and

$$\mathbb{P}(|S_n - \mu| \ge t) \le 2\exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Proof. Define $Y_i = X_i - \mathbb{E}[X_i]$, so $\mathbb{E}[Y_i] = 0$ and $a_i - \mathbb{E}[X_i] \le Y_i \le b_i - \mathbb{E}[X_i]$. The width of the interval for Y_i is still $b_i - a_i$. Also, $S_n - \mu = \sum_{i=1}^n Y_i$.

For any $s > 0$, by the Chernoff method (Proposition 7.3):

$$\mathbb{P}(S_n - \mu \ge t) = \mathbb{P}\left(e^{s(S_n-\mu)} \ge e^{st}\right) \le e^{-st}\,\mathbb{E}\left[e^{s\sum_{i=1}^n Y_i}\right].$$

Since the Y_i are independent,

$$\mathbb{E}\left[e^{s\sum_i Y_i}\right] = \prod_{i=1}^n \mathbb{E}[e^{sY_i}].$$

By Hoeffding's lemma (Lemma 7.4), each factor satisfies $\mathbb{E}[e^{sY_i}] \le \exp(s^2(b_i-a_i)^2/8)$. Therefore

$$\mathbb{P}(S_n - \mu \ge t) \le e^{-st}\prod_{i=1}^n \exp\left(\frac{s^2(b_i-a_i)^2}{8}\right) = \exp\left(-st + \frac{s^2}{8}\sum_{i=1}^n (b_i-a_i)^2\right).$$

Minimizing over $s > 0$: the exponent $h(s) = -st + \frac{s^2}{8}\sum_i (b_i - a_i)^2$ is a quadratic in s with minimum at

$$s^* = \frac{4t}{\sum_{i=1}^n (b_i-a_i)^2}.$$

Substituting:

$$h(s^*) = -\frac{4t^2}{\sum_i (b_i-a_i)^2} + \frac{16t^2}{8(\sum_i (b_i-a_i)^2)} = -\frac{4t^2}{\sum_i (b_i-a_i)^2} + \frac{2t^2}{\sum_i (b_i-a_i)^2} = -\frac{2t^2}{\sum_i (b_i-a_i)^2}.$$

This gives the one-sided bound. For the two-sided bound, apply the same argument to $-Y_i$ (which has the same width) to get $\mathbb{P}(S_n - \mu \le -t) \le \exp(-2t^2/\sum_i (b_i-a_i)^2)$, then use the union bound. □

Corollary 7.6 (Hoeffding for i.i.d. Bounded Variables)**.** *If $X_1, \ldots, X_n$ are i.i.d. with $a \le X_i \le b$ and $\mu = \mathbb{E}[X_1]$, then for the sample mean $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$,*

$$\mathbb{P}(|\bar{X}_n - \mu| \ge \varepsilon) \le 2\exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right).$$

Proof. Apply Theorem 7.5 to $S_n = n\bar{X}_n$ with $t = n\varepsilon$ and $b_i - a_i = b - a$ for all i:

$$\mathbb{P}(|n\bar{X}_n - n\mu| \ge n\varepsilon) \le 2\exp\left(-\frac{2n^2\varepsilon^2}{n(b-a)^2}\right) = 2\exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right).$$ □

7.5 Sample Complexity via Hoeffding

Proposition 7.7 (Hoeffding Sample Complexity)**.** *Let $X_1, \ldots, X_n$ be i.i.d. with $a \le X_i \le b$. To ensure $\mathbb{P}(|\bar{X}_n - \mu| \ge \varepsilon) \le \delta$, it suffices to take*

$$n \ge \frac{(b-a)^2}{2\varepsilon^2}\ln\frac{2}{\delta}.$$

Proof. By Corollary 7.6, we need $2\exp(-2n\varepsilon^2/(b-a)^2) \le \delta$. Solving:

$$\exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \le \frac{\delta}{2}, \quad \text{so} \quad \frac{2n\varepsilon^2}{(b-a)^2} \ge \ln\frac{2}{\delta}, \quad \text{i.e.,} \quad n \ge \frac{(b-a)^2}{2\varepsilon^2}\ln\frac{2}{\delta}.$$ □

Example 7.8 (Hoeffding vs. Chebyshev Sample Size)**.** Consider the same setting as Example 7.2: i.i.d. Bernoulli rewards with $a = 0$, $b = 1$, $\varepsilon = 0.05$, and $\delta = 0.05$. Hoeffding requires

$$n \ge \frac{(1-0)^2}{2\cdot 0.05^2}\ln\frac{2}{0.05} = \frac{1}{0.005}\ln(40) = 200\cdot 3.689 \approx 738.$$

Compare with Chebyshev's requirement of $n \ge 2000$. Hoeffding gives a nearly 2.7× reduction in sample size. This improvement comes from exploiting the bounded range of the variable, not just its variance.

Sample Complexity: Chebyshev vs. Hoeffding

For estimating the mean of bounded i.i.d. random variables $X_i \in [a, b]$ within accuracy ε and confidence $1 - \delta$:

Method	Sample complexity	Dependence on δ
Chebyshev	$n \geq \sigma^2/(\varepsilon^2\delta)$	Polynomial: $O(1/\delta)$
Hoeffding	$n \geq \frac{(b-a)^2}{2\varepsilon^2} \ln(2/\delta)$	Logarithmic: $O(\ln(1/\delta))$

For high-confidence estimates (small δ), Hoeffding's logarithmic dependence on $1/\delta$ vastly outperforms Chebyshev's polynomial dependence. This is why exponential tail bounds are essential in PAC learning theory.

Monte Carlo Policy Evaluation Sample Complexity

Consider estimating $V^\pi(s)$ by averaging n i.i.d. returns $G^{(1)}, \ldots, G^{(n)}$. If rewards are bounded in $[0, R_{\max}]$ and the discount factor is $\gamma < 1$, each return satisfies $0 \leq G^{(i)} \leq R_{\max}/(1-\gamma)$. By Proposition 7.7,

$$n \geq \frac{R_{\max}^2}{2(1-\gamma)^2\varepsilon^2} \ln \frac{2}{\delta}$$

episodes suffice to ensure $|\bar{G}_n - V^\pi(s)| < \varepsilon$ with probability at least $1-\delta$. This is the basic sample complexity of Monte Carlo policy evaluation. Note the dependence on $(1-\gamma)^{-2}$: as the discount factor approaches 1, the effective horizon grows and far more episodes are needed.

8 Exercises

Exercise 8.1 (MGF Computations). $(*)$
Let $X \sim \text{Geometric}(p)$ with $\mathbb{P}(X = k) = (1-p)^{k-1}p$ for $k = 1, 2, \ldots$

(a) Compute the MGF $M_X(t)$ and determine the values of t for which it is finite.
(b) Use the MGF to compute $\mathbb{E}[X]$ and $\text{Var}(X)$.

Exercise 8.2 (Jensen's Inequality Applications). $(*)$
Let X be a positive random variable with finite expectation.

(a) Use Jensen's inequality with $g(x) = 1/x$ (which is convex on $(0, \infty)$) to prove that $\mathbb{E}[1/X] \geq 1/\mathbb{E}[X]$.
(b) Use Jensen with $g(x) = -\ln x$ to prove that $\mathbb{E}[\ln X] \leq \ln(\mathbb{E}[X])$.
(c) Let $X \sim \text{Uniform}(1, 3)$. Verify part (a) numerically by computing both $\mathbb{E}[1/X]$ and $1/\mathbb{E}[X]$.

Exercise 8.3 (Chebyshev Sharpness). $(**)$
Construct a random variable X with $\mathbb{E}[X] = 0$ and $\text{Var}(X) = 1$ such that $\mathbb{P}(|X| \geq 2) = 1/4$, showing that Chebyshev's bound $\mathbb{P}(|X| \geq 2) \leq 1/4$ is tight for $k = 2$.
Hint: Consider a three-point distribution.

Exercise 8.4 (WLLN Does Not Require Identical Distributions). $(**)$
Let $X_1, X_2, \ldots$ be independent (but not necessarily identically distributed) random variables with $\mathbb{E}[X_i] = \mu$ for all i and $\text{Var}(X_i) \leq C$ for some constant $C < \infty$. Use Chebyshev's inequality to prove that $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i \xrightarrow{\mathbb{P}} \mu$.

Exercise 8.5 (Monte Carlo Sample Complexity). (∗∗)
An RL agent estimates $V^\pi(s)$ using Monte Carlo rollouts. Rewards are bounded in $[0, 1]$ and $\gamma = 0.99$.

(a) Compute the range $[0, R_{\max}/(1-\gamma)]$ of a single discounted return.
(b) Using Hoeffding's inequality, how many episodes are needed to ensure the estimate is within $\varepsilon = 0.5$ of the true value with probability at least 0.95?
(c) How does the answer change if $\gamma = 0.9$?
(d) Compare the Hoeffding sample complexity with the Chebyshev sample complexity, using the bound $\mathrm{Var}(G) \le (b-a)^2/4$.

Exercise 8.6 (Proving Hoeffding via Symmetrization). (∗ ∗ ∗)
Let X be a random variable with $\mathbb{E}[X] = 0$ and $a \le X \le b$. An alternative route to Hoeffding's lemma uses the following steps:

(a) Let X' be an independent copy of X. Show that $\mathbb{E}[e^{t(X-X')}] \le \mathbb{E}[e^{tX}]^2$.
Hint: Use Jensen's inequality applied to the conditional expectation $\mathbb{E}[e^{-tX'} \mid X]$.
(b) The symmetric variable $X - X'$ satisfies $|X - X'| \le b - a$. Using the fact that a symmetric bounded random variable Y with $|Y| \le c$ satisfies $\mathbb{E}[e^{tY}] \le e^{t^2c^2/2}$ (which can be shown via the Taylor series of cosh), conclude that $\mathbb{E}[e^{tX}] \le e^{t^2(b-a)^2/8}$.

9 Summary

- The *moment generating function* $M_X(t) = \mathbb{E}[e^{tX}]$ encodes all moments via $\mathbb{E}[X^n] = M_X^{(n)}(0)$ and, when it exists in a neighborhood of the origin, uniquely determines the distribution.
- The MGF of a sum of independent random variables factorizes: $M_{X+Y}(t) = M_X(t)M_Y(t)$. This immediately shows that sums of independent Poissons are Poisson and sums of independent normals are normal.
- *Markov's inequality* $\mathbb{P}(X \ge a) \le \mathbb{E}[X]/a$ requires only non-negativity and a finite first moment.
- *Chebyshev's inequality* $\mathbb{P}(|X - \mu| \ge k) \le \sigma^2/k^2$ uses the first two moments and gives polynomial tail decay.
- *Cauchy-Schwarz* $|\mathbb{E}[XY]|^2 \le \mathbb{E}[X^2]\mathbb{E}[Y^2]$ and *Jensen's inequality* $g(\mathbb{E}[X]) \le \mathbb{E}[g(X)]$ (for convex g) are workhorse tools throughout probability and optimization.
- The *weak law of large numbers* states $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$ (proved via Chebyshev); the *strong law* strengthens this to $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$ (proof deferred to Phase 01).
- The *central limit theorem* states $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1)$: sample means are approximately normal for large n, regardless of the underlying distribution.
- *Hoeffding's inequality* provides exponential concentration for bounded random variables: $\mathbb{P}(|\bar{X}_n - \mu| \ge \varepsilon) \le 2e^{-2n\varepsilon^2/(b-a)^2}$.
- The Hoeffding sample complexity $n \ge \frac{(b-a)^2}{2\varepsilon^2} \ln(2/\delta)$ depends logarithmically on $1/\delta$, vastly outperforming Chebyshev's polynomial dependence for high-confidence estimation.

As the capstone lesson of the Probability subdomain, we now review the arc of Lessons 9–12.

Lesson	Title	Core Contribution
9	Probability Spaces and Conditioning	Kolmogorov axioms, conditional probability, Bayes' theorem, independence. Provides the foundational language.
10	Random Variables and Distributions	Random variables, PMF/PDF/CDF, Bernoulli, Binomial, Geometric, Poisson, Uniform, Exponential, Normal distributions. Introduces the objects we compute with.
11	Expectation, Variance, and Joint Distributions	Linearity of expectation, LOTUS, variance, covariance, joint and conditional distributions, tower property, law of total variance. Provides the computational machinery.
12	Inequalities and Limit Theorems	MGFs, Markov/Chebyshev/Cauchy-Schwarz/Jensen inequalities, WLLN, CLT, Hoeffding's inequality, sample complexity bounds. Provides the asymptotic and finite-sample guarantees.

10 Further Reading

- **Phase 01, Lesson 4** (Measure Spaces) provides the measure-theoretic foundations that make the definitions in Lessons 9–12 fully rigorous, replacing finite sums and integrals with Lebesgue integration.
- **Phase 01, Lesson 5** (Probability Foundations) revisits convergence concepts, proves the strong law of large numbers using the Borel-Cantelli lemma, establishes the CLT via characteristic functions, and develops martingale theory.
- **Phase 01, Lesson 5** also covers the *Azuma-Hoeffding inequality* for martingale difference sequences, which generalizes Hoeffding's inequality to dependent random variables – essential for analyzing online RL algorithms where successive observations are correlated.
- **Phase 02** (Core RL) uses the law of large numbers to justify Monte Carlo estimation, Chebyshev and Hoeffding inequalities for finite-sample convergence bounds, and the CLT for asymptotic confidence intervals in policy evaluation.
- For a comprehensive treatment of concentration inequalities beyond Hoeffding (including Bernstein, Bennett, and McDiarmid's inequalities), see Durrett [2019, Chapter 2] and Grimmett and Stirzaker [2020, Appendix A].
- The Chernoff bound technique (optimizing over t in Markov's inequality applied to e^{tX}) is a special case of the *exponential tilting* method, which plays a central role in large deviations theory; see Feller [1968, Chapter 8] for an introduction.
- Historically, the law of large numbers was first proved by Jakob Bernoulli (1713) in a special case (Bernoulli trials). Chebyshev (1867) gave the first general proof using his inequality. The central limit theorem was established by Lindeberg (1922) and Feller (1935) under progressively weaker conditions.

References

Joseph K. Blitzstein and Jessica Hwang. *Introduction to Probability.* CRC Press, 2nd edition, 2019.

Rick Durrett. *Probability: Theory and Examples.* Cambridge University Press, 5th edition, 2019.

William Feller. *An Introduction to Probability Theory and Its Applications*, volume 1. Wiley, 3rd edition, 1968.

Geoffrey Grimmett and David Stirzaker. *Probability and Random Processes.* Oxford University Press, 4th edition, 2020.

Sheldon M. Ross. *A First Course in Probability.* Pearson, 10th edition, 2019.